Performance Analysis of Shunt Active Power Filter based on PIDA Controller

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***Abstract*—This research paper addresses current harmonics mitigation using shunt active power filter. Due to excessive use of power electronic converters (PECs) and non-linear loads it is observed that power system voltage and current responses become non sinusoidal and thus produce waveform driven power quality issues like voltage harmonic and current harmonic. The active power filters are gained attention to mitigate the harmonics, power factor correction and reactive power compensation. Three phase reference source current is determined with active reactive power (P-Q) theory with PIDA (Proportional Integral Derivative Acceleration) controller as DC bus voltage regulation. PIDA controller is an extended form of proportional integral derivative (PID) controller. The extra term 'A' define for acceleration with this new phrase a closed loop network can respond quicker (less settling time) with lesser overshoot. The proposed system is simulated using MATLAB/Simulink. An effective compensation solution with defined harmonic limit is obtained to improve power quality as per electrical regulatory commission and standards such as IEEE 519-1992.**

Keywords-Proportional Integral Derivative Acceleration (PIDA); Shunt Active Power Filter (SAPF); Active Reactive Power theory (P-Q); Capacitor Coupled Voltage Source Converter (CC-VSC).

I. INTRODUCTION

With the prompt growth of non-linear loads as diode, PECs loads are the main sources of harmonics and reactive power which influence performance indices of power system network [1-2]. The pollution of electrical power is severe at the utilization level and is caused by more losses in transmission & distribution network, interference problem in communication systems, transformer overheating, neutral burning, nuisance tripping of protective device and at time result in operation failure of electronic apparatus because it contains micro and nano electronic based controller system which works with very minimum power level. It is conformed that non-sinusoidal current outcome for the utility power supply company like poor power factor, low power efficiency, distortion of line voltage, electromagnetic interference (EMI) etc.

The harmonics can be filtered out using a series of power filters such as passive, active and hybrid [3]. It can be connected in series, shunt or combination of these two at point of common coupling (PCC) depending upon nature of load. The passive filters have disadvantage as ageing, largeness, risk of resonance, fixed compensation and lossy circuitry [4].

To overcome the issue of power quality, research have been made on active power filtering. The active power filters (APFs) have gained interest because of awesome performance to mitigate the voltage current harmonic, power factor correction and reactive power compensation (KVAR) [5]. Active power filters has been used for both voltage and current harmonics mitigation with reactive power compensation [8]. The performance of shunt active filter depends upon control techniques [6-7]. Various time-domain based control topologies like instantaneous active and reactive power (P-Q) [9-10], instantaneous active and reactive current theory (I_d-I_q), synchronous reference frame (SRF) [11], sliding mode control [17], self tuning filter [12], positive sequence detection using phase locked loop (PLL) [13] are present in literature to get reference source current.

Selection of decorous reference compensation current taking out scheme acting vital role in APF performance. So, the proposed work discusses the voltage source inverter (VSI) topology for SAPF with control strategy based on instantaneous active and reactive power (P-Q) method. It has simplicity of calculations, which involves only algebraic calculation. PIDA controller is proposed to regulate DC bus voltage for switching loss compensation. This controller delivers faster and smoother response as compared to other conventional controller [14-16].

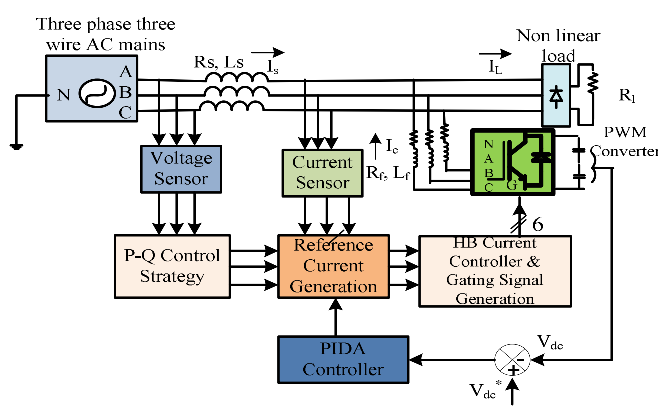

Figure 1. Proposed system layout.

II. SYSTEM CONFIGURATION

Proposed system architecture is shown in Fig.1. It consist three-phase supply source connected with uncontrolled converter feeding resistance (R_l) load. Uncontrolled (Diode bridge) converter produces the non-linearity into the utility system by drawing current harmonics and more reactive

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power. A current controlled voltage source converter (CC-VSC) is used as an active power filter which is connected at PCC in shunt with non-linear load. The IGBT (SW1, SW3, SW5) conducts for positive half cycle and IGBT (SW2, SW4, SW6) conducts for negative half cycle. Shunt APF cancels harmonic current component at utility side and makes supply current almost in-phase with source voltage by providing reactive power compensation. The shunt APF is connected at PCC via coupling or commutation inductors (L_f, R_f). Three smoothing inductors (Ls, Rs) are connected in series with non-linear load. A DC-link capacitor C_{dc} is connected at the inverter DC side for achieving self support DC bus. The DC bus voltage is measured and compared with reference DC link voltage.

III. Reference Filter Current Generation

The block diagram for whole (P-Q) theory has been shown as Fig 2.

$$\begin{bmatrix} I_{l\alpha} \\ I_{l\beta} \end{bmatrix} = \sqrt{2/3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} I_{la} \\ I_{lb} \\ I_{lc} \end{bmatrix} \tag{1}$$

$$\begin{bmatrix} V_{s\alpha} \\ V_{s\beta} \end{bmatrix} = \sqrt{2/3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} V_{sa} \\ V_{sb} \\ V_{sc} \end{bmatrix} \tag{2}$$

The instantaneous values of active power (P), reactive power (Q) in three phase circuit is expressed as:

$$\begin{bmatrix} P \\ Q \end{bmatrix} = \begin{bmatrix} V_{s\alpha} & V_{s\beta} \\ -V_{s\beta} & V_{s\alpha} \end{bmatrix} \begin{bmatrix} I_{l\alpha} \\ I_{l\beta} \end{bmatrix} \tag{3}$$

Capacitor leaking and switching losses are also present. Hence, extra average power is demanded to mitigate for losses going on inside VSC due to on-off of semiconductor devices is described in equation (4).

$$\Delta P = \bar{P}_{loss} + \tilde{P} \tag{4}$$

The reference filter compensating current components can be evaluated from instantaneous reactive power (Q) and instantaneous real power harmonic component (ΔP).

$$\begin{bmatrix} I_{f\alpha}^* \\ I_{f\beta}^* \end{bmatrix} = \frac{1}{V_{s\alpha}^2 + V_{s\beta}^2} \begin{bmatrix} V_{s\alpha} & V_{s\beta} \\ -V_{s\beta} & V_{s\alpha} \end{bmatrix} \begin{bmatrix} \Delta P \\ Q \end{bmatrix} \tag{5}$$

After all, reference filter currents in three phase three-wire (3P-3W) can be achieved as following equations.

$$\begin{bmatrix} I_{fa}^* \\ I_{fb}^* \\ I_{fc}^* \end{bmatrix} = \sqrt{2/3} \begin{bmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \\ -1/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} I_{f\alpha}^* \\ I_{f\beta}^* \end{bmatrix} \tag{6}$$

IV. PIDA Controller

Where, $\bar{P}_{loss}$ is the average loss taking place inside VSI, which is carried from self supported DC-bus voltage regulator as per (7). It is intended to give good compensation and superb transient response. True DC-bus voltage (V_{dc}) is compared with the reference (V_{dc}*) and error ΔV_{dc} is managed in PIDA controller.

$$\bar{P}_{loss} = K_P \Delta V_{dc} + K_I \int \Delta V_{dc}.dt + K_D \frac{d\Delta V_{dc}}{dt} + K_A \frac{d^2 \Delta V_{dc}}{dt^2} \tag{7}$$

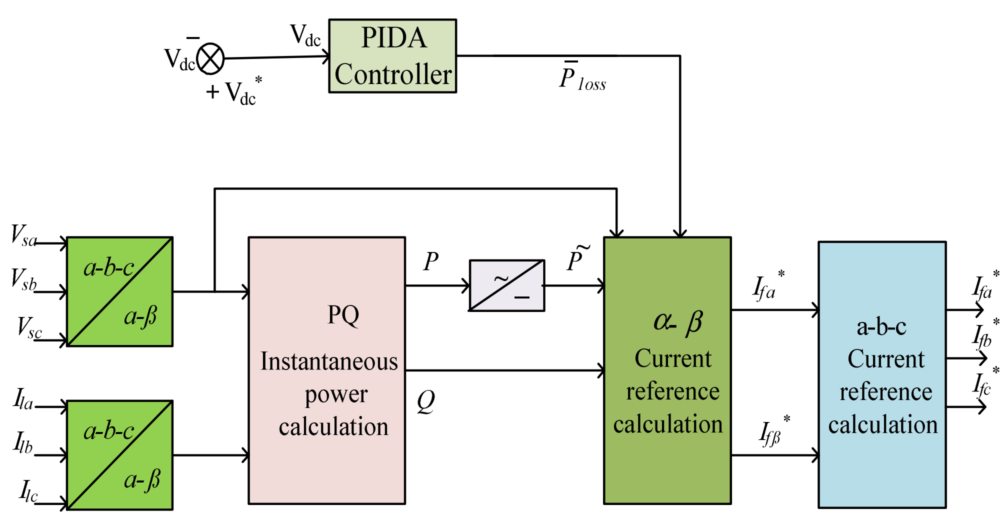

Figure 2. Control block diagram for P – Q theory.

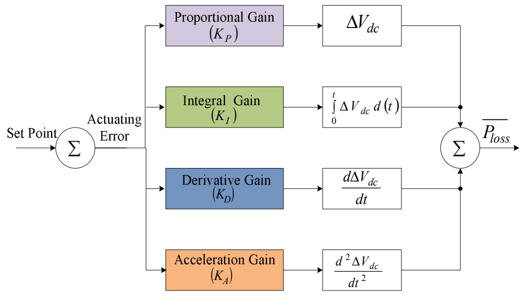

Figure 3. PIDA controller.

V. SIMULATION RESULTS

Network framework has been sketched in Fig. 1. It has been realized in MATLAB/Simulink (R2015a) so as to demonstrate its performance with PIDA controller. A designed proposed system is used a low pass filter (LPF) as 2nd order butterworth filter. Table 1 illustrates system design parameters and specification for MATLAB-simulation study.

TABLE I. SYSTEM DESIGN DATA FOR MATLAB-SIMULINK MODELING

Source	Fundamental supply voltage 400 V (rms phase-phase) Source frequency = 50 Hz
APF	DC- bus capacitor, C_{dc1}, C_{dc2}, = 40µF Reference DC-bus voltage, V_{dc}^*=850 V Filter resistor and inductor, L_f = 3.35 mH, R_f = 0.4 Ω PIDA controller gain K_P = .000628 and K_I = .00628, K_D=.1768*10^{-13}, K_A==.1768*10^{-14}
Loads	Three phase uncontrolled rectifier feeding with R_l = 10 Ω load

Prior to 0.01second due to non-linear loads harmonic are brought in network, supply current turn into non linear. At 0.01 sec., Shunt APF comes in role and compensated source current response is created.

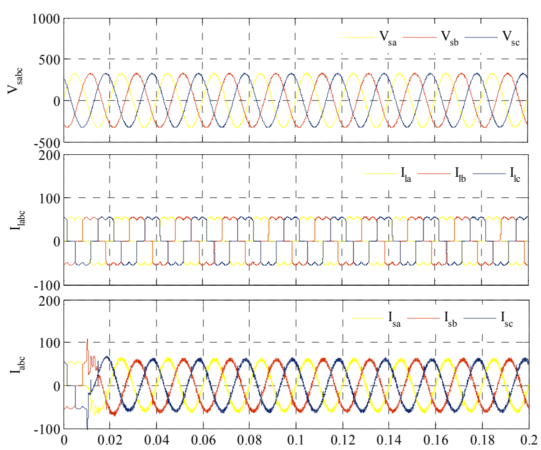

Figure 4. Source current (V_{sa}), load current (I_{la}) and source current (I_{sa}).

Performance parameter source current (V_{sa}), load current (I_{la}), and supply current (I_{sa}) are represented in Fig. 4 from up to down using PIDA controller under nonlinear load condition. Three phase filter current is illustrated in Fig. 5. It is zero before .01.

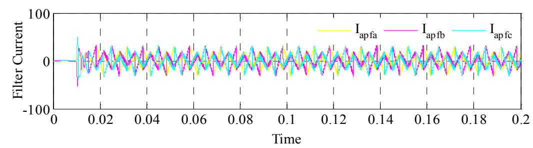

Figure 5. Three phase filter current.

DC bus voltage (V_{dc}) with PIDA controller is given away in Fig. 6. The DC bus capacitor voltage achieves steady-state value of its reference in few cycles.

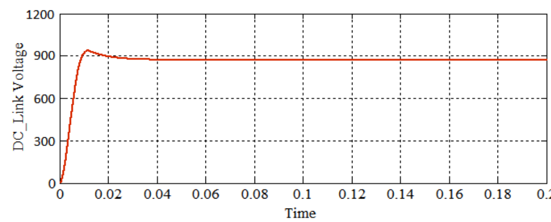

Figure 6. DC link voltage.

It is evaluated that the input power factor is improved from 0.85 to 0.99 as depicted in Fig. 7.

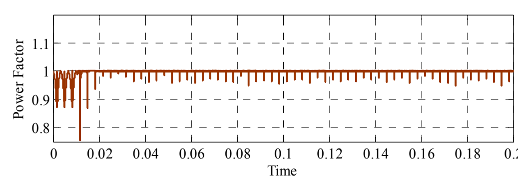

Figure 7. Power factor.

Active and reactive power (P-Q) level is shown in Fig. 8. Active power increment and compensation in reactive power show improvement in power quality.

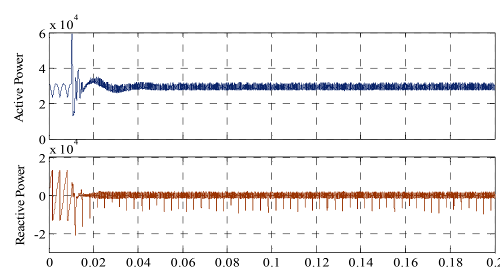

Figure 8. Active and reactive power.

The harmonic spectrum of uncompensated and compensated source current of phase ‘A’ ‘B’ and phase ‘C’ is depicted in Fig. 9 from up to down respectively. After compensation, source currents THD reduce from 30.32% to 2.77% for phase ‘A’. This shows a significant reduction in

THD for steady-state condition of load. It is found that the supply current THDs are well within the recommended limits.

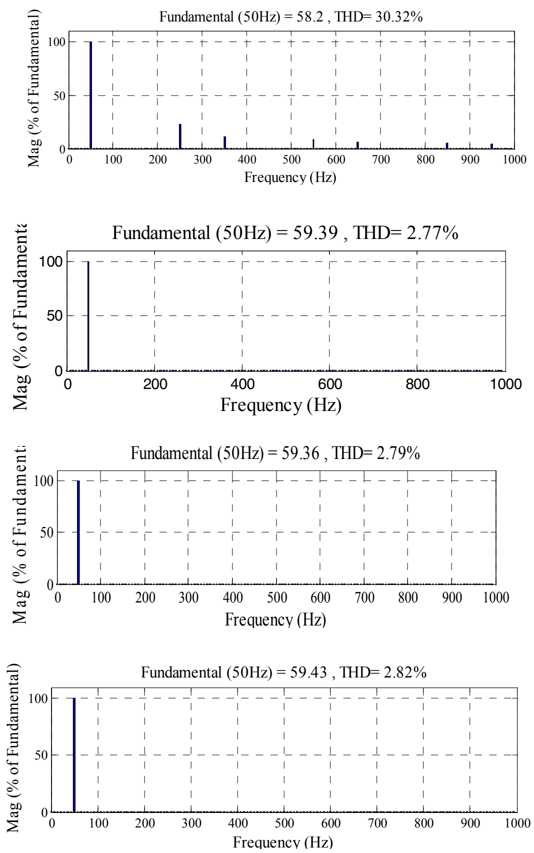

Figure 9. Spectral analysis of uncompensated and compensated supply current for 'A' 'B' and 'C' phase respectively.

VI. CONCLUSION

This article has been examined reference supply current taking out for 3-phase 3- wire shunt APF using instantaneous active and reactive power (P-Q) theory based control scheme for power quality improvement. Voltage controller such as PIDA has been implemented to regulate DC bus capacitor voltage of capacitor coupled voltage source converter (CC-VSC). The system performance with proposed controller has been assessed in form of harmonic mitigation and DC bus voltage regulation. It composes supply current sinusoidal thus reducing its total harmonic distortion (THD) level well below 5% (IEEE 519-1992 standards) and concurrently makes the supply current in phase with the supply voltage with approximate unity power factor (.99).

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